

Problem 10

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Problem 1 For the piecewise function f defined by

$$f(x) = \begin{cases} \sin(x) & \text{if } x \leq 0 \\ \frac{x^2}{x^2 - 4} & \text{if } 0 < x < 2 \\ \frac{x^2}{x^2 + 4} & \text{if } 2 \leq x \end{cases}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Candidates for vertical asymptote: $x^2 - 4 = 0 \implies x = -2, x = 2$ (Each piece of f is continuous at all other points).

Note: $x = -2$ is not a vertical asymptote, since $f(x) = \sin(x)$ near -2 , so $\lim_{x \rightarrow -2} f(x) = \lim_{x \rightarrow -2} \sin(x) = \sin(-2)$.

$$\lim_{x \rightarrow 2^-} \frac{x^2}{x^2 - 4} = \lim_{x \rightarrow 2^-} \frac{x^2}{(x - 2)(x + 2)} = -\infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, and the denominator negative.

The line $x = 2$ is a vertical asymptote of f because $\lim_{x \rightarrow 2^-} f(x) = -\infty$. This is the only vertical asymptote of f .

- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Check End Behavior as $x \rightarrow \infty$:

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$\lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 4}$ is of the form: $\frac{\infty}{\infty}$

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 4} &= \lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 4} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} \\ &= \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{4}{x^2}} \\ &= 1\end{aligned}$$

The line $y = 1$ is a horizontal asymptote of f since $\lim_{x \rightarrow \infty} f(x) = 1$.

Check End Behavior as $x \rightarrow -\infty$:

$\lim_{x \rightarrow -\infty} \sin(x)$ does not exist, since the function oscillates.

For example, $f(-n\pi) = \sin(-n\pi) = 0$, while $f(-(2n+1)\frac{\pi}{2}) = \sin(-(2n+1)\frac{\pi}{2}) = 1$, for all positive integers n .

There is no horizontal asymptote as $x \rightarrow -\infty$ since $\lim_{x \rightarrow -\infty} f(x)$ does not exist.